B. Nikita and Books

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, math, sortings | https://codeforces.com/problemset/problem/2244/B

As is well known, Nikita loves reading. Today, he made a mess in his room and arranged his books into n stacks in a row, numbered from 1 to n from left to right. The i-th stack contains a_i books. This arrangement is called *neat* if, in every stack except the rightmost one, the number of books is strictly less than in the stack to its right; that is, the array a is strictly increasing.

Yura wants to make the arrangement *neat* by performing the following operation any number of times:

1. Choose a stack i such that 1 ≤ i lt n and a_i gt 1.
2. Take 1 book from the top of stack i, so a_i decreases by 1.
3. Put this book on top of stack i + 1, so a_{i+1} increases by 1.

Determine whether Yura can make the arrangement *neat*.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the number of stacks.

The second line of each test case contains n integers a_i ($1 \le a_i \le 10^9$) — the initial number of books in each stack.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output "YES" if Yura can make the arrangement *neat*, and "NO" otherwise.

You may output each letter in any case (lowercase or uppercase). For example, the strings "yEs", "yes", "Yes", and "YES" will be accepted.

Example

Input
7 3 1 2 3 3 1 1 2 3 10 1 1 3 2 2 2 4 1 4 2 2 5 8 2 8 1 8 4 1 1 3 5

Output
YES NO YES YES NO YES NO